

PROB & STATS: Assignment 02 Date: _____

Answer#01:

$$f(x, y) = 24xy, \quad 0 \leq x, y \leq 1, \quad x + y \leq 1$$

X = Turkish

Y = domestic

$$a. \quad P(x > 1/2) = \int_{1/2}^1 \int_0^{1-x} 24xy \, dy \, dx$$

$$= \int_{1/2}^1 24x \int_0^{1-x} y \, dy \, dx$$

$$= \int_{1/2}^1 24x \left[\frac{y^2}{2} \right]_0^{1-x} dx$$

$$= \int_{1/2}^1 24x \frac{(1-x)^2}{2} dx$$

$$= \int_{1/2}^1 (12x - 24x^2 + 12x^3) dx$$

$$= \left[6x^2 - 8x^3 + 3x^4 \right]_{1/2}^1$$

$$P(x > 1/2) = 6 - 8 + 3 - \left[6(1/4) - 8(1/8) + 3(1/16) \right]$$

$$P(x > 1/2) = \frac{5}{16}$$

$$P(x > 1/2) = 0.3125 \quad \text{Ans.}$$

$$b. f_y(y) = \int_0^{1-y} f(x,y) dx$$

$$= \int_0^{1-y} 24xy dx$$

$$= 24y \int_0^{1-y} x dx$$

$$= 12y [x^2]_0^{1-y}$$

$$= 12y (1-y)^2$$

$$f_y(y) = 12y - 24y^2 + 12y^3$$

Ans.

$$c. P(x < 1/8 \mid y = 3/4) = \int_0^{1/8} f(x \mid y = 3/4) dx \div \int_0^{1-y} f(x,y) dx$$

$$= \int_0^{1/8} 24x(3/4) dx \div (12y[1-y]^2)$$

$$= \int_0^{1/8} 18x dx \div [12(3/4)[1-3/4]^2]$$

$$= \frac{18}{2} \left[x^2 \right]_0^{1/8} \div \frac{9}{16}$$

$$= \frac{18}{2} (1/8)^2 \times \frac{16}{9}$$

$$P(x < 1/8 \mid y = 3/4) = 0.25 \quad \text{Ans.}$$

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Answer #02:

$$f(x,y) = \begin{cases} ye^{-y(1+x)} & , x,y \geq 0 \\ 0, & \text{elsewhere} \end{cases}$$

a.

$$f_x(x) = \int_0^{\infty} f(x,y) dy = \int_0^{\infty} ye^{-y(1+x)} dy$$

$$f_x(x) = \frac{1}{(1+x)^2} \quad \text{Ans.}$$

$$f_y(y) = \int_0^{\infty} f(x,y) dx$$

$$= \int_0^{\infty} ye^{-y(1+x)} dx = ye^{-y} \int_0^{\infty} e^{-yx} dx$$

$$= ye^{-y} \cdot \frac{1}{y}$$

$$f_y(y) = e^{-y} \quad \text{Ans.}$$

$$b. P(x > 2, y > 2) = \int_2^{\infty} \int_2^{\infty} f(x,y) dx dy$$

$$= \int_2^{\infty} \int_2^{\infty} ye^{-y(1+x)} dx dy$$

$$= e^{-y} \int_2^{\infty} e^{-y(1+x)} dx = \int_2^{\infty} e^{-3y} dy$$

$$= \frac{e^{-6}}{3}$$

$$P(x > 2, y > 2) = 0.00083 \quad \text{Ans.}$$

Answer #03:

$$f(x,y) = \begin{cases} \frac{3x-y}{9} & , 1 < x < 3, 1 < y < 2 \\ 0 & , \text{elsewhere} \end{cases}$$

$$\begin{aligned} \text{a. } f_x(x) &= \int_1^2 \frac{3x-y}{9} dy = \frac{3x}{9} \int_1^2 dy - \frac{1}{9} \int_1^2 y dy \\ &= \frac{3x}{9} |y|_1^2 - \frac{|y^2|_1^2}{18} \\ &= \frac{x}{3} (1) - \frac{(4-1)}{18} \end{aligned}$$

$$f_x(x) = \frac{x}{3} - \frac{1}{6} \quad \text{Ans.}$$

$$\begin{aligned} f_y(y) &= \int_1^3 \frac{3x-y}{9} dx = \frac{1}{9} \int_1^3 x dx - \frac{y}{9} \int_1^3 dx \\ &= \frac{|x^2|_1^3}{6} - \frac{y}{9} |x|_1^3 \\ &= (8/6) - 2y/9 \end{aligned}$$

$$f_y(y) = \frac{4}{3} - \frac{2y}{9} \quad \text{Ans.}$$

b. $f(x,y) = f_x(x) f_y(y)$, if independent.

$$f_x(x) = \frac{2x-1}{6} , \quad f_y(y) = \frac{12-2y}{9}$$

L.H.S $\neq$ R.H.S

→ Also prove by inserting values $x = 1.5, y = 1.25$

$$\frac{3(1.5)-1.25}{9} \neq \left[\frac{2(1.5)-1}{6} \right] \left[\frac{12-2(1.25)}{9} \right]$$

0.361 $\neq$ 0.351 $\Rightarrow$ Hence X, Y are NOT Independent

$$\begin{aligned}
 c. \quad P(x > 2) &= \int_2^3 f_x(x) dx \\
 &= \int_2^3 \left(\frac{2x-1}{6} \right) dx \\
 &= \frac{1}{6} \int_2^3 2x dx - \frac{1}{6} \int_2^3 dx \\
 &= \frac{1}{3} \left[x^2 \right]_2^3 - \frac{1}{6} [x]_2^3 \\
 &= \frac{1}{3} (9-4) - \frac{1}{6} (3-2)
 \end{aligned}$$

$$P(x > 2) = 2/3 \quad \text{Ans.}$$

Answer #04:

a. PMF_x : ?

$$P(x=1) = 0.4 - 0 = 0.4$$

$$P(x=3) = 0.6 - 0.4 = 0.2$$

$$P(x=5) = 0.8 - 0.6 = 0.2$$

$$P(x=7) = 1.0 - 0.8 = 0.2$$

b. $P(4 < x \leq 7)$: ?

$$\Rightarrow = P(x=5) + P(x=6) + P(x=7)$$

$$P(4 < x \leq 7) = 0.2 + 0 + 0.2 = 0.4 \quad \text{Ans.}$$

Answer # 05:

$$f(x, y) = \frac{9}{16} \cdot \frac{1}{4^{x+y}}, \text{ for } x = 0, 1, 2, \dots, y = 0, 1, 2, \dots$$

$$a. f_x(x) = \frac{9}{16} \cdot \frac{1}{4^x} \sum_{y=0}^{\infty} \frac{1}{4^y}$$

$$= \frac{9}{16} \cdot \frac{1}{4^x} \cdot \frac{4}{3}$$

$$f_x(x) = \frac{3}{4 \cdot 4^x}$$

$$f_y(y) = \frac{9}{16} \cdot \frac{1}{4^y} \sum_{x=0}^{\infty} \frac{1}{4^x}$$

$$= \frac{9}{16} \cdot \frac{1}{4^y} \cdot \frac{4}{3}$$

$$f_y(y) = \frac{3}{4 \cdot 4^y}$$

$$f(x, y) = f_x(x) \cdot f_y(y)$$

$$\frac{9}{16} \cdot \frac{1}{4^{x+y}} = \left(\frac{3}{4 \cdot 4^x} \right) \left(\frac{3}{4 \cdot 4^y} \right) \Rightarrow \text{L.H.S} = \text{R.H.S}$$

Hence, they are independent

$$b. P(x+y < 4) = \sum_{x=0}^3 \sum_{y=0}^{3-x} f(x, y)$$

$$\Rightarrow = \sum_{x=0}^3 \sum_{y=0}^{3-x} \left(\frac{3}{4 \cdot 4^x} \right) \left(\frac{3}{4 \cdot 4^y} \right) = \frac{9}{16} \sum_{x=0}^3 \sum_{y=0}^{3-x} \frac{1}{4^{x+y}}$$

$$= \frac{63}{64}$$

$$P(x+y < 4) = 0.99 \quad \text{Ans.}$$

Answer #06:

$$f(y) = \begin{cases} 10(1-y)^9, & 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

- a. i. $f(y) \geq 0$ for all values of y
 $\Rightarrow 10(1-y)^9$ is always non-negative
as $0 \leq y \leq 1$
 $\rightarrow$ First condition is satisfied.

ii. $\int_y f(y) = 1$, $\int_0^1 f(y) dy = \int_0^1 10(1-y)^9 dy$
 $= \left[-(1-y)^{10} \right]_0^1 = (-0)^{10} - (-1)^{10} = 1$
 $\rightarrow$ Second condition satisfied.

b. $P(0.6 < y \leq 1) = \int_{0.6}^1 f(y) dy$
 $= \int_{0.6}^1 10(1-y)^9 dy$
 $= \left[-(1-y)^{10} \right]_{0.6}^1$

$$= -(0)^{10} - (-(1-0.6)^{10})$$
$$P(0.6 < y \leq 1) = (0.4)^{10} \quad \text{Ans.}$$

Answer #07:

$$E(x) = n \frac{k}{N} = 3 \times \frac{1}{5} = 0.6 \quad \text{Ans.}$$

$$\begin{aligned} \text{Var}(x) &= n \frac{k}{N} \times \frac{N-k}{N} \times \frac{(N-n)}{(n-1)} \\ &= \frac{3 \times 1}{5} \times \frac{4}{5} \times \frac{2}{4} \end{aligned}$$

$$\text{var}(x) = 0.24 \quad \text{Ans.}$$

Answer #08:

$$f(x, y) = \begin{cases} \frac{16y}{x^3}, & x > 2, 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

 $\int xy$: ?

$$\begin{aligned} E(x^2) &= \int \int_{xy} x^2 f(x, y) = \int_2^{\infty} \int_0^1 x^2 \cdot \frac{16y}{x^3} dy dx = \int_2^{\infty} \int_0^1 \frac{16y}{x} dy dx \\ &= \int_2^{\infty} \left[\frac{8y^2}{x} \right]_0^1 dx = \int_2^{\infty} \frac{8}{x} dx = [8 \ln x]_2^{\infty} \\ &= 8 \ln \infty - 8 \ln 2 \\ &= \infty \end{aligned}$$

$\Rightarrow$ The mean of (x^2) is not finite, so
variance of x does not exist,
thus we cannot compute the correlation
coefficient.

Answer #09:

$$a. P(X=x) = \binom{n}{x} p^x (1-p)^{n-x} = \binom{8}{x} 0.6^x (0.4)^{8-x}$$

$$P(X=3) = \binom{8}{3} (0.6)^3 (0.4)^5$$

$$P(X=3) = 0.12386 \quad \text{Ans.}$$

b. If atleast 5 took for non-psychological problems, then at most 3 took for psychological.

$$P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$$

$$= (0.4)^8 + \binom{8}{1} (0.6)^1 (0.4)^7$$

$$+ \binom{8}{2} (0.6)^2 (0.4)^6$$

$$+ \binom{8}{3} (0.6)^3 (0.4)^5$$

$$P(X \leq 3) = 0.1737 \quad \text{Ans.}$$

Answer #10:

$$a. P(X=x) = \frac{e^{-\lambda} (\lambda)^x}{x!}$$

$$P(X=5) = \frac{e^{-3} (3)^5}{5!} = 0.1008 \quad \text{Ans.}$$

$$b. P(X < 3) = P(X=0) + P(X=1) + P(X=2)$$

$$= e^{-3} \left(\frac{3^0}{0!} + \frac{3^1}{1!} + \frac{3^2}{2!} \right)$$

$$P(X < 3) = 0.4232 \quad \text{Ans.}$$

$$\begin{aligned}
 c. \quad P(x \geq 2) &= 1 - P(x \leq 1) \\
 &= 1 - P(x=0) - P(x=1) \\
 &= 1 - \frac{e^{-3} \cdot 3^0}{0!} - \frac{e^{-3} \cdot 3^1}{1!}
 \end{aligned}$$

$$P(x \geq 2) = 0.8008 \quad \text{Ans.}$$

Answer # 11:

a. Since, mean time lies in the range when it is desirable, this shows that the resin performance is desirable more than it is undesirable. Ans.

$$b. \quad z = \frac{x - \mu}{\sigma}$$

$$z_1 = \frac{1-3}{0.5} = -4, \quad z_2 = \frac{4-3}{0.5} = 2$$

$$\begin{aligned}
 P(1 < x < 4) &= P(-4 < z < 2) \\
 &= P(z < 2) - P(z < -4) \\
 &= 0.9772 - 0.000032
 \end{aligned}$$

$$P(1 < x < 4) = 0.977168 \quad \text{Ans.}$$

$$\begin{aligned}
 P(\text{undesirable}) &= 1 - p(1 < x < 4) \\
 &= 1 - 0.977168 \\
 &= 0.0228 \quad \text{or} \quad 2.28\% \quad \text{Ans.}
 \end{aligned}$$

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Answer # 12:

$$P(x < 95) = P(x < 94.5), \quad z = \frac{x - \mu}{\sigma}$$

$$P(x < 95) = P(z < -1.71)$$

$$P(x < 95) = 0.0436 \quad z = \frac{94.5 - 115}{12}$$

$$z = -1.71$$

$$E(x) = np = 600 \times 0.0436$$

$$E(x) = 26.16$$

$$E(x) \approx 26 \text{ students}$$

Ans.